Overview of Detectif.ai Voice Security

https://gemini.google.com/app/f3da2fdf4bf9187d

User prompt: tell me all abt https://detectif.ai/

Response: **Detectif.ai** is an AI voice security, authentication, and call intelligence platform built to protect organizations—such as banks, lenders, insurers, and contact centers—against voice cloning, deepfakes, account takeovers, and fraudulent impersonations over telephone channels.

Core Models and Capabilities

Detectif.ai offers four specialized models designed for real-time speech analysis and telephone infrastructure:

1. **Aletheia (Deepfake Detection):**
    - Designed to detect synthetic, cloned, or replayed speech on live calls or recordings.
    - Tailored specifically for telephone channels (e.g., 8 kHz, G.711 μ-law passband), analyzing spectral representations, pitch, formants, breath noise, and vocoder artifacts.
    - Noted for its high speed and accuracy (benchmarked with ultra-low commercial latency and high detection rates).
2. **Eidos (Speaker Verification):**
    - Compares a caller's voice against an enrolled reference voiceprint to return a speaker match score.
    - Used alongside deepfake detection to prevent cases where an attacker might match a voiceprint using a high-quality clone.
3. **Noesis (Call Intelligence):**
    - Classifies the recipient or other side of a call at the very beginning (distinguishing between humans, IVRs, voicemails, or voice agents) to enable adaptive call routing.
4. **Lexis (Voice Agents):**
    - Manages policy-gated conversational workflows, executing authentication checks before allowing sensitive actions or transferring conversations and evidence records to human specialists.

Use Cases in High-Risk Environments

- **Contact Center Account Takeovers:** Screens inbound calls before a customer can request account changes, password resets, or phone number updates using a cloned voice.
- **Payment Authorization & Impersonation:** Detects and flags high-risk instructions or urgent transfer requests made by fake executives, relatives, or impersonated customers.

- **Remote Onboarding & Identity Verification:** Combines deepfake detection and liveness/speaker checks during digital account opening.

Deployment & Integration

Detectif.ai supports flexible deployment topologies:

- **Cloud, Private VPC, On-Premise, or On-Device:** Includes INT8 handset builds for local mobile app execution.
- **Telephony & Agent Integration:** Integrates with SIP/RTP media forks, contact center audio streams, and developer frameworks like LiveKit and Pipecat for real-time pipeline monitoring.

User prompt: let's say these people are conductin a hackathon... what would be a great idea forit ?

Response: A hackathon hosted by Detectif.ai would perfectly bridge the gap between AI innovation and high-stakes security. Given their focus on telephony, voice agents, and deepfake detection, a brilliant theme would be **"VoiceArmor: Defending the Future of FinTech and Decentralized Wealth."**

This would challenge developers to build next-generation voice security layers for high-value financial platforms, focusing on scenarios where a single cloned voice could lead to massive losses.

Here is how the hackathon could be structured, featuring specific tracks that utilize Detectif.ai's core models:

1. The High-Stakes Trading Vault (Eidos + Aletheia)

The Challenge: Build an ultra-secure voice authorization layer for equity trading or cryptocurrency exchange platforms. **The Concept:** Participants would design an integration where executing a high-value stock trade or crypto transfer triggers a mandatory voice prompt. The system must use **Eidos** to verify the user's voiceprint and **Aletheia** to ensure it isn't a deepfake or a high-quality replay attack. **The Edge:** Developers who can optimize the integration to run locally on a handset —using low-level programming like C to deploy Detectif's INT8 on-device models for zero-latency checks—would score major technical points.

2. Prompt Engineering the "Honeypot" Agent (Lexis)

The Challenge: Design an intelligent customer service voice agent that actively traps scammers. **The Concept:** Instead of just hanging up on a deepfake, teams would use prompt engineering and the **Lexis** model to create a dynamic AI voice agent. If the agent detects a synthetic voice trying to reset a password or change a registered phone number, the AI seamlessly shifts its system prompt to play along. It would gather threat intelligence, waste the attacker's time, and silently alert the fraud team, all without letting the scammer know they've been caught.

3. The Startup / Executive Impersonation Defense (Noesis)

The Challenge: Stop the "fake CEO" or "fake VC" scam in real-time. **The Concept:** Scammers are increasingly using deepfakes to impersonate startup founders or executives instructing finance teams to wire funds. Teams would build a conference-call screening tool that uses **Noesis** to classify whether a participant is a real human, an IVR, or a synthetic voice agent, instantly throwing up a red flag on the team's dashboard if a synthetic voice enters the startup pitch or board meeting.

The Finale: To cap it off, the hackathon could conclude with a high-energy startup mixer or open-mic pitch session, where teams have to demo their platform live by actively trying to "hack" their own voice-security systems on stage using off-the-shelf voice cloning tools, proving that their Detectif.ai integration holds the line.

What kind of track would you want to build in a setup like this—something more focused on the core AI models, or building the front-end user experience?

User prompt: more niche ideas from internet ?/ gemme some articles research papers abt this

Response: Based on recent research papers from arXiv and datasets like ASVspoof 5, the academic and technical community is heavily focused on the gap between clean "studio-quality" deepfake detection and messy "in-the-wild" telephone environments.

If you want to host a highly technical, research-grounded hackathon track around Detectif.ai, here are some niche, cutting-edge ideas based on current challenges in synthetic speech detection:

1. The "Partial Deepfake" Telephony Challenge

The Research Context: A February 2025 paper (arXiv:2502.03559) highlights that lower layers of Self-Supervised Learning (SSL) models are best at catching deepfakes, but current systems struggle immensely with **partial deepfakes**—where a scammer splices a few seconds of a synthetic voice into a mostly authentic conversation (e.g., just synthesizing the word "Yes" or a specific account number). **The Hackathon Idea:** Build a system that can detect micro-injections of synthetic speech over a simulated 8kHz G.711 μ-law telephone line.

- **The Task:** Teams must use Detectif.ai's API to scan a 3-minute customer service call and accurately flag a single 2-second synthesized sentence hidden among real audio, without triggering false positives on the low-quality telephony codec artifacts.

2. The Adversarial "Wash" Defense

The Research Context: The latest ASVspoof 5 dataset (released late 2024/early 2025) introduced adversarial attacks to audio deepfakes for the first time. Attackers are now using tools to intentionally add background noise (like traffic or cafe sounds) or intentionally compress the audio to mask the "vocoder artifacts" (the unnatural harmonic smoothness) that deepfake detectors rely on. **The Hackathon Idea:** The "Adversarial Clean-Room" challenge.

- **The Task:** Participants act as the "blue team." They are fed highly compressed, aggressively degraded deepfake audio designed specifically to bypass detectors. They must build a preprocessing pipeline (using noise-reduction, frequency isolation, etc.) that "cleans" the audio just enough for Detectif.ai's **Aletheia** model to catch the synthetic vocoder artifacts without warping the audio so much that it causes a false positive.

3. The Cross-Lingual Identity Heist

The Research Context: Current research indicates that many deepfake detectors are over-trained on English speech. When scammers clone a voice but make the synthetic voice speak in Spanish, Mandarin, or Hindi, detection accuracy can drop. **The Hackathon Idea:** A cross-lingual voice verification challenge using **Eidos**.

- **The Task:** A user enrolls their voiceprint in English. The challenge is to build an authentication gateway that can accurately accept the legitimate user speaking in a second language (e.g., Hindi), while blocking a high-quality deepfake clone that is also speaking Hindi.

These ideas shift the focus from basic app-building to tackling the actual cryptographic and acoustic challenges that companies like Detectif.ai are actively trying to solve in the real world.
